Department of Mechanical Engineering

Mechanical Engineering Design

ENME 301 Assignment 2: Bearing and Housing

Cover Sheet for online submission

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Introduction

A “Horizontal-Shaft Skydive Simulator” is to be designed for a new Christchurch based location. The initial Simulator design consists of a horizontal fan driven by a rotating shaft powered by an already available diesel engine through a universal drive shaft. A rough sketch was given for an initial design concept and is shown in Figure 1.

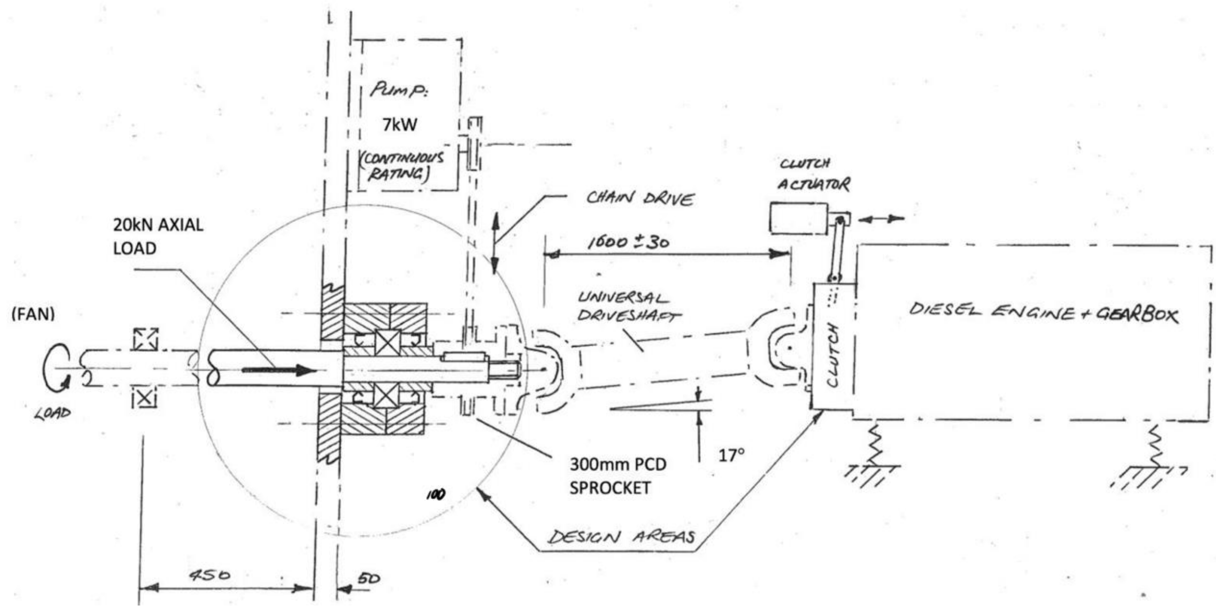

Figure 1: Initial Design Sketch [1]

The shaft is supported radially by two bearings. The inner bearing (furthest from the fan) also supports the shaft axially and is the primary focus of this design report. This report covers the selection process for an axial bearing as well as a design of the bearing housing.

A sprocket located between the bearing and the universal drive shaft transmits power through a chain to a hydraulic pump which actuates the variable pitch system for the fan. The shaft-sprocket power transmission is considered in this report, as well as connection of the shaft to the universal driveshaft joint. A clutch connects the diesel engine to the other end of the driveshaft and is responsible for making sure the shaft is not overpowered. The shaft design and clutch analysis are also detailed in this report.

Loads

Before considering design options, all loads and moments on the shaft must be determined. A free body diagram of the shaft from the fan to the sprocket is shown in Appendix A and Table 1 presents a summary of the calculated values for this figure.

Table 1: Summary of Loads in Fan-Sprocket Shaft Section.

Load	Symbol	Value (N)	Location x (m)	Direction
Fan axial force	F_{fan}	20,000	0	➡
Fan weight force	W_{fan}	1,226	0	⬇
Outer bearing y reaction force	R_{outer_y}	2,320	0.4	⬆
Inner bearing y reaction force	R_{inner_y}	1,837	0.9055	⬇
Inner bearing x reaction force	R_{inner_x}	20,000	0.9055	⬅
Chain force	F_{chain}	743	1.06	⬆

Bending moment and shear force diagrams for the shaft are presented in Appendix A.

The bending moment diagram shows the primary bending moments of the shaft due to the radial shear forces imparted on it by the weight of the fan and tension of the sprocket driven chain. These moments occur in the xy plane. Secondary moments in the xz plane must also be considered due to the universal driveshaft being at some angle to the horizontal.

The shear force diagram shows a shear force and bending moment diagram along the xz plane of the shaft. As the forces induced by driveshaft are oscillating, only the maximum shear and bending moments are considered in the diagram. The direction of the moments is arbitrary as the forces oscillate between maximums in either orientation. A summary table of primary, secondary, and combined bending moments along the shaft section is presented in Table 2.

Table 2: Bearing Reaction Moments Summary Table

Support	**Primary Moment (Nm) - xy plane**	**Abs Max Secondary Moment (Nm) - xz plane**	**Abs Max Combined Moment (Nm)**
Outer Bearing	-490	0	490
Inner Bearing	115	2676	2678

The torque along the shaft also must be considered for shaft dimension calculations. At its maximum continuous rating (MCR), the $550\ kW$ diesel engine imparts a torque of $8754\ Nm$ on the shaft rotating at $600\ RPM$. At its maximum capacity, the hydraulic pump absorbs $7\ kW$ of power from the main shaft and the torque on the sprocket is calculated to be $111\ Nm$. However, all shaft calculations assume a maximum torque of $8754 Nm$ along the entire length of the shaft as if the pump were producing no power or the chain was not attached.

Shaft Calculations

Shaft Diameter

The material selected for manufacturing the main shaft is AISI 1050 carbon steel with a yield strength of $580\ MPa$ and an ultimate tensile strength of $690\ MPa$ [2]. Using the American Society of Mechanical Engineers (ASME) shaft diameter formula [3], a minimum acceptable diameter was calculated for the 1050 steel. Bending moment and torsional moment factors were chosen by selecting the upper bound of the range of values for a suddenly applied load with minor shocks allowed. The minimum required diameter for a Factor of Safety (FOS) of 2.0 was determined to be $105\ mm$. This was rounded up to $110\ mm$ to fit a standard SKF bearing dimensions giving a FOS of 2.3. An alternative 'reality check' was conducted on this minimum shaft diameter using the maximum shear stress theory of failure, and the FOS was found to be 4.3. All shaft diameter calculations were done considering the shaft operating at its minimum critical radius (MCR) and having its maximum torsional moment throughout the entire shaft (as if the chain were removed from the sprocket).

Once the spline which connects to the universal driveshaft had been specified, a further FOS was calculated using the minimum diameter of the spline. With a diameter of $92\ mm$, the FOS was determined with the ASME equation to be 1.4.

Key Calculations

The material of the key to transmit power from the shaft to the sprocket was chosen to be hot rolled AISI 1006 low carbon steel with a yield strength of $165\ MPa$ [4]. This is a weaker steel than what is used for the shaft and sprocket, as the key is designed to fail before these components. The key has a rectangular cross-sectional area with a height of $9\ mm$ and a width of $12\ mm$, and it inserts into the shaft by half of its height, $4.5\ mm$. It was specified that they key would fail at 2 times the pumps maximum operating torque. Calculations for shear and compressive failure of the key at this FOS gave a minimum required length of $11\ mm$. At a normal operating torque, the key has a FOS of 6.2 in shear and 2.0 in compression. The key was designed to fail in compression if the hydraulic pump is overloaded by more than twice its maximum capacity to protect it from being damaged further and protecting the shaft and sprocket from damage.

Shaft Features

To axially restrain the shaft to the inner race of the bearing, a shoulder was incorporated into the shaft's design. This introduced a stress concentration factor into the shaft which needed to be accounted for in safety calculations. Figure 4 in Appendix A shows the stress concentration diagrams for shaft shoulders in both bending and torsion.

With a minimum shoulder diameter of $122\ mm$, as specified by SKF [5], and a fillet radius of $3\ mm$, the stress concentration factors were calculated to be 2.23 and 1.55 for bending and torsion respectively. The FOS for bending at the shaft shoulder was calculated to be 12.7. Assuming the shaft is operating at its MCR, and the pump is not removing power from the shaft, the FOS in torsion is 6.5. This was calculated using the Von Mises criterion for the max shear stress of 1050 steel. These FOS values are acceptable for the shaft shoulder.

Bearing Selection

Bearing Selection and Alternatives

There are several suitable options for bearing types that would be suitable for this application. Using Appendix C [5], bearings could be chosen that can facilitate both radial and axial loads. This narrowed down the possible selection to bearing types 4, 5, 6, 7, 8 and 11.

Type 5, a spherical roller thrust bearing was chosen to explore first because the ratio of axial to radial load the bearing would be experiencing is heavily biased towards the axial load. However, bearing life calculations soon discovered that due to the large size of the bearing required to fit around the shaft, the axial load is relatively small compared to the capacity of the bearing, and that an axially biased bearing may not be required. Therefore, light to medium load bearings and even more radially biased bearings were explored next.

Preliminary bearing housing designs also drew attention to the need for a bearing that can take axial loads in both directions. This property axially constrains the bearing assembly even when the fan is turned off. This requirement also invalidated our initial selection of a type 5 bearing. Appendix D shows which bearings can support an axial load in both directions and using both Appendix C and D the bearing selection could be narrowed down to types 7, 8 and 9. The next bearing explored was type 7, a four-point contact ball bearing, because it is the most space efficient of the three types. Bearing life calculations confirmed that it is a suitable choice.

Bearing Life Calculations

The bearing life was calculated in Appendix A. The radial and axial load factors used in the calculations for a four-point contact ball bearing with a predominantly axial load are 0.6 and 1.07 respectively [5]. The results of these calculations, based on an 8 hour day 300 days a year, can be seen in Table 3.

Table 3: Bearing life calculations results.

	Million Revolutions	**Hours**	**Years**
L_{10}	1911	53079	22.12

Lubrication

LGMT-2 grease was selected because it is a general-purpose industrial grease with good mechanical stability. It is a mineral oil based, lithium soap thickened grease suitable for a wide range of applications including agricultural equipment and industrial fans. The lubrication condition of the bearing (K) was calculated to be 1.83 in Appendix A. This falls within the typical $1-4$ range and confirms that LGMT-2 is a suitable lubricant.

Clutch Selection

A clutch connecting the gearbox to the universal driveshaft is to be recommended for the simulator. The clutch needs to act as an overload slipping mechanism and must fit within the existing gearbox housing of width $450\ mm$. To fit within these constraints, an outer clutch plate diameter of $440\ mm$ is recommended. Due to the size limits, a multidisc clutch can be used. The clutch is specified to slip at 5% more than the torque output of the motor at its MCR. Using a multidisc clutch with 4 pairs of cast iron on cast iron friction surfaces, the actuation force required to keep the clutch engaged until the slipping torque was calculated assuming uniform wear of the friction surfaces. The actuation force required against the clutch's inner diameter was graphed for lubricated, greased and dry plates and is presented in Figure 2 (left). For the same clutch and friction factors, the maximum surface contact pressure was graphed against inner diameter and is shown in Figure 2 (right).

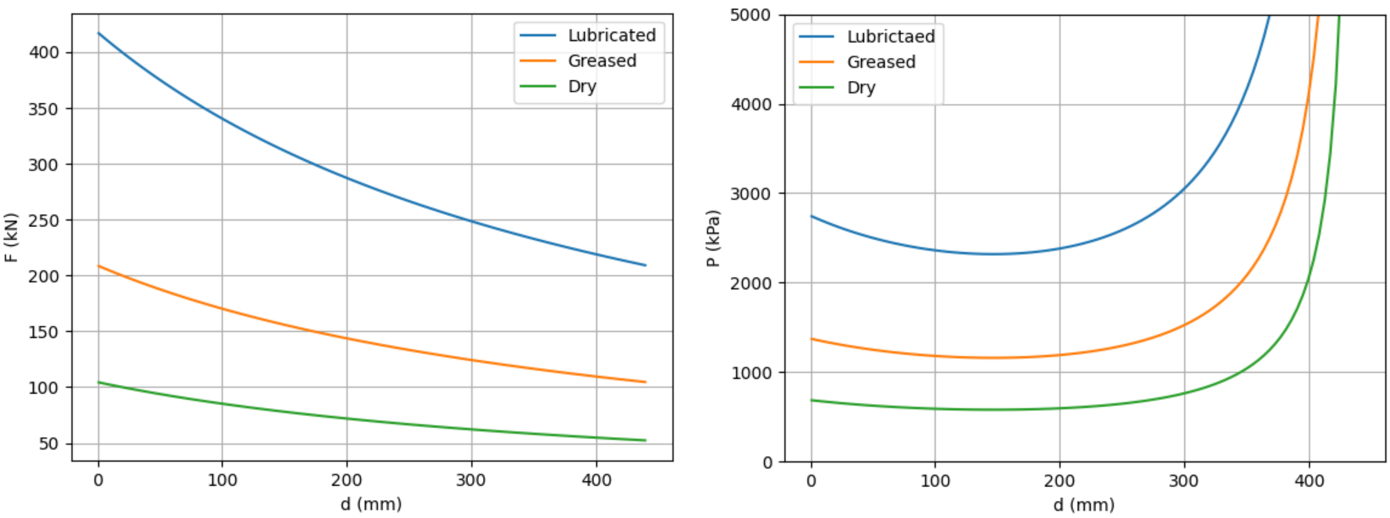

Figure 2: Actuation Force vs Clutch Inner Diameter (left) and Maximum Contact Pressure vs Clutch Inner Diameter (right) for Different Friction Factors.

Figure 2 indicates that the maximum contact pressure is at a minimum when the inner diameter of the shaft is around $150\ mm$. To balance between actuation force and the life of the clutch, a greased surface is recommended. Table 4 shows the full clutch recommendation for the simulator.

Table 4: Recommended Clutch Specifications.

Outer Diameter (mm)	Inner Diameter (mm)	No. of Plate Pairs	Materials	Surface Conditions
440	150	4	Cast-Iron on Cast-Iron	Greased

This clutch has a maximum actuation force of $156\ kN$ per plate and a maximum surface contact pressure of $1.16\ MPa$.

Seals

Seal Selection

Radial shaft seals sit between two components in relative motion, ideal for sealing between the shaft and bearing housing in this case. A standard radial seal was selected from SKF because they are suitable for general industrial applications and are metric standard, giving an ideal fit to our metric shaft [6]. A 130x160x12 HMS5 RG was chosen.

Contact Surface Specification

The seal counterface should have a hardness of at least $55\ HRC$ to achieve a reliable seal and good service life. The surface roughness of the counterface is also important and should be between 0.2 and $0.8\ \mu m$ for rotary applications such as this [7]. Based on these conditions the bare shaft is not suitable and two seal counterfaces / spacers are required.

440C is a high carbon martensitic stainless steel with good hardness and wear resistance, high strength and moderate corrosion resistance. After heat treatment 440C has the highest hardness, wear resistance and strength of any stainless-steel alloy. 440F has the same hardness and hardenability as 440C but has better machineability [8]. Therefore, the seal counterfaces were specified to be made from 440F stainless steel with a surface finish of $0.5\mu m$.

Housing and Overall Design

The requirements of the bearing assembly are to firmly secure the bearing and sprocket in place while protecting the bearing from debris and providing access for lubrication. It should be simple and easy to assemble, disassemble and maintain.

Design

The lower race of the bearing is held in place by a shoulder on the inside and a lock nut on the outside. From inside to outside there is a spacer / seal counterface up against the shoulder, followed by the bearing, another counterface and the sprocket. A lock nut on the outside of the sprocket clamps everything together against the shoulder.

The upper race of the bearing is secured in place by clamping two halves of the bearing housing together using four M16 bolts. The same four bolts also secure the housing to the wall and are complete with flat and spring washers to prevent loosening during operation. The housing also holds the seals in the correct locations against their counterfaces and provide cavities for lubricant. The lubricant can be pumped in through a nipple in the top of the outer housing. It fills up the outer cavity, flows through the bearing into the inner cavity and finally exits through a hole in the inner housing. When grease comes out through the hole the housing is full and the hole can be plugged with a grub screw.

Assembly / Disassembly

First a counterface is slid onto the shaft up against the shoulder followed by the inner housing with the seal already mounted in it. Then the bearing can be slid on followed by the second seal counterface and outer housing also with the seal already mounted in it. The key can then be placed in the keyway and the sprocket slid on over it until it is up against the outer counterface. Finaly a lock washer and a lock nut can be threaded onto the end to clamp everything against the shoulder, ensuring that the lock washer fits into the keyway and that one of the washer tabs has been folded into one of the grooves on the lock nut. The same process in reverse can be used during disassembly.

Housing and Fixture Strength

The fixtures holding the housing to the wall are four M16 grade A bolts that need to withstand the $20\ kN$ axial load of the fan. Additionally, if the bolts are not tightened correctly then they will need to withstand a shear force caused by the radial load on the bearing.

Calculations were also done to evaluate the strength of the housing at its smallest cross sections, again in both the axial and radial directions. The results of these calculations can be seen in Table 5.

Table 5: Results of housing and fixture strength calculations.

	Bolt Shearing	**Bolt Stripping**	**Housing Bearing and Housing Shear Failure**
FOS	43.3	21.2	$\gg 100$

Conclusion

The design of the Horizontal-Shaft Skydive Simulator integrates components to ensure efficient and reliable operation. Design decisions included selecting a four-point contact ball bearing for handling both radial and axial loads, resulting in a bearing life over 22 years, and designing the main shaft with a diameter of $110\ mm$, providing a Factor of Safety (FOS) of 2.3. The key material, AISI 1006 low carbon steel, was chosen to ensure it fails before the shaft or sprocket, protecting these components from damage. The bearing housing was designed for secure bearing placement, debris protection, and easy maintenance. A multidisc clutch, designed to slip at 5% above the motor's maximum continuous rating torque with greased cast-iron surfaces, was recommended to balance actuation force and clutch life. Radial shaft seals were selected to fit the metric shaft, with 440F stainless steel specified for the seal counterfaces to ensure hardness and wear resistance. The bearing housing and its fixtures were designed to withstand the axial load of the fan and potential shear forces, ensuring structural integrity. This analysis and careful selection of materials, dimensions, and components ensure the simulator will operate efficiently and safely, providing a good foundation for its implementation in Christchurch.

References

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[2] AZO Materials, “AISI 1050 Carbon Steel (UNS G10500),” 2024. [Online]. Available: https://www.azom.com/article.aspx?ArticleID=6526. [Accessed 2024].

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[5] SKF, “Rolling Bearings,” 2018. [Online]. Available: https://cdn.skfmediahub.skf.com/api/public/0901d196802809de/pdf_preview_medium/0901d196802809de_pdf_preview_medium.pdf. [Accessed 2024].

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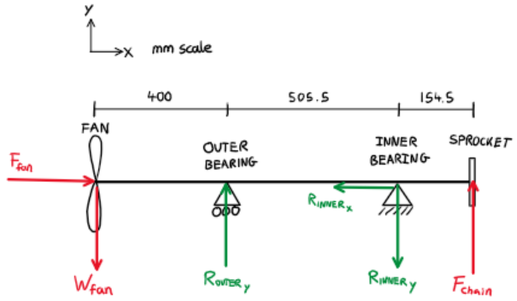
[7] SKF, “Design of Associated Components,” [Online]. Available: https://www.skf.com/au/products/industrial-seals/power-transmission-seals/radial-shaft-seals/low-cross-sectional-height/design-associated-components. [Accessed 2024].

[8] Penn Stainless Products, “440C STAINLESS STEEL,” 2024. [Online]. Available: https://www.pennstainless.com/resources/product-information/stainless-grades/400-series/440c-stainless-steel/. [Accessed 2024].

Appendix A (Hand Calculations)

FBD & Loads / Moments Calcs
(using final dimensions)

FBD:



- $F_{fan} = 20{,}000\ N$
- $W_{fan} = 125_{kg} \times 9.81_{ms^{-2}} = 1226\ N$
- $F_{chain} = 743\ N$

} Known forces

Reaction force calculations:

$$\Sigma F_x = 0 = F_{fan} - R_{INNER_x}$$

$$\boxed{\therefore R_{INNER_x} = 20{,}000N \leftarrow}$$

$$\Sigma M_{(INNER\ BEARING)} = 0 = W_{fan} \times 0.9055 - R_{OUTER_y} \times 0.5055 + F_{chain} \times 0.1545$$

$$0.5055\ R_{OUTER} = 1226 \times 0.9055 + 743 \times 0.1545$$

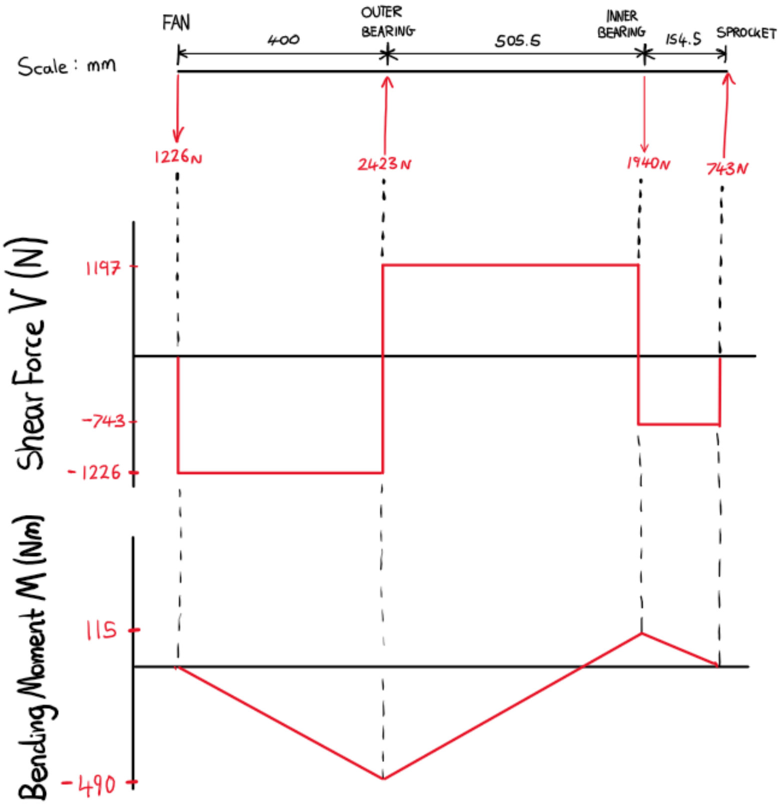
$$\boxed{\therefore R_{OUTER_y} = 2423\ N \uparrow}$$

$$\Sigma M_{(OUTER\ BEARING)} = 0 = W_{fan} \times 0.4 + R_{INNER_y} \times 0.5055 + F_{chain} \times 0.66$$

$$0.5055\ R_{INNER_y} = -1226 \times 0.4 - 743 \times 0.66$$

$$\therefore R_{INNER_y} = -1940N$$

$$\boxed{R_{INNER_y} = 1940N \downarrow}$$



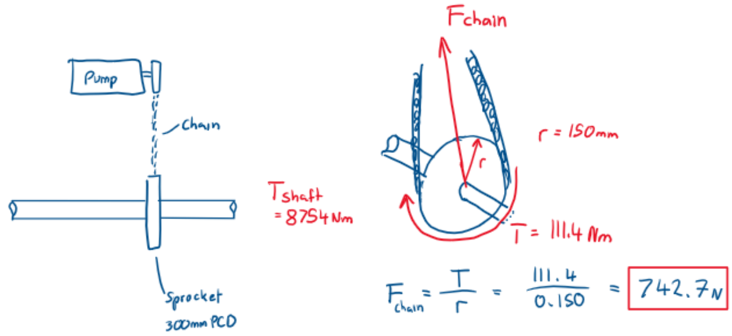
$$M_{(OUTER\ BEARING)} = -1226 \times 0.4 = \underline{-490Nm}$$

$$M_{(INNER\ BEARING)} = -490 + 1197 \times 0.5055 = \underline{115\ Nm}$$

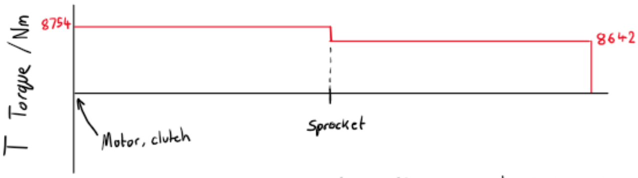
Pump/Sprocket Calcs

Torque required by sprocket:

$$T = \frac{P}{\Omega} = 7 \times 10^3 \times \frac{60}{600 \times 2\pi} = 111.4\ Nm$$



$$F_{chain} = \frac{T}{r} = \frac{111.4}{0.150} = \boxed{742.7N}$$



Secondary Momement Calculations

$$M_{dl} = 8754Nm$$

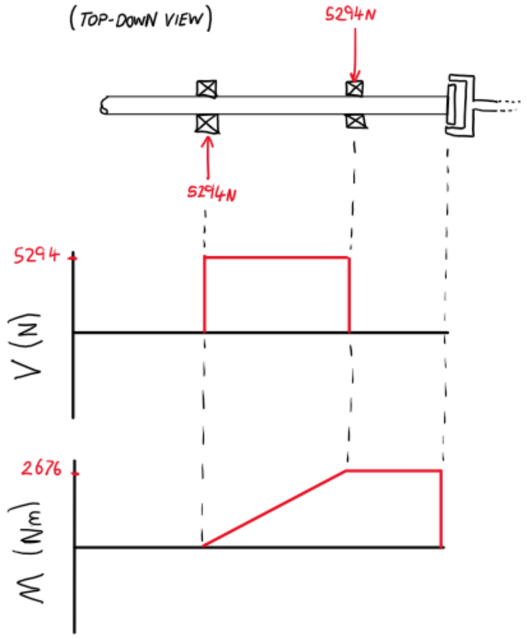
$$\beta = 17°$$

$$a = 0.5055$$

$$A_{max} = B_{max} = \frac{M_{dl} \cdot \tan\beta}{a} = \frac{8754 \times \tan 17°}{0.5055}$$

$$= 5294\ N$$

(TOP-DOWN VIEW)



$$M_{(INNER\ BEARING)} = 5294 \times 0.5055 = 2676\ Nm$$

Maximum bending moment at inner bearing occurs when the secondary moments are greatest.

$$M_{INNER}(max) = \sqrt{\underbrace{M_{INNER_{(xy)}}^2}_{primary\ moment} + \underbrace{M_{INNER_{(xz)}}^2}_{secondary\ moment}}$$

$$= \sqrt{115^2 + 2676^2}$$

$$M_{INNER}(max) = 2678\ Nm$$

Shaft Calculations

\ Torque created by motor at MCR:

Motor generates 550 kW, shaft spins at 600 RPM

$$T = \frac{P}{\Omega} = 550\times10^3 \times \frac{60}{600\times2\pi} = 8754 \text{ Nm}$$

Maximum moment of shaft = 2678 Nm (at inner bearing)

ASME Diameter calculation:

Load constants:

$C_m = 2.0$

$C_t = 1.5$

from upper values of 'Load applied suddenly - minor shocks' range

$$d = \left\{ \frac{5.1n}{\tau_p}\left[(C_m M)^2 + (C_t T)^2\right]^{1/2} \right\}^{1/3}$$

n = FOS

$$d = \left\{ \frac{5.1n}{124\times10^6}\left[(2.0\times2678)^2 + (1.5\times8754)^2\right]^{1/2} \right\}^{1/3}$$

$$n = \frac{d^3 \tau_p}{5.1\left[(C_m M)^2 + (C_t T)^2\right]^{1/2}}$$

$\boxed{d = 83.6 \text{ mm} \quad (n=1, \text{ critical diameter})}$

$d = 95.6$ mm $(n = 1.5)$

$d = 105.3$ mm $(n = 2)$

→ Main shaft minimum diameter chosen to be 110mm:

$\boxed{d = 110\text{mm} \rightarrow n = 2.3}$

Material Selection

$\boxed{\text{AISI 1050 Carbon Steel}}$

$\sigma_y = 580$ MPa $\quad \sigma_{UTS} = 690$ MPa

$\tau_p = 0.3\sigma_y$ or $0.18\sigma_{UTS}$ whichever smallest

$0.3 \times 580 = 174$ MPa

$0.18 \times 690 = 124$ MPa smallest

$\therefore \tau_p = 124$ MPa

Alternative Reality Checks:

Using first principles for determining the maximum shear stress:

$$\tau_{max} = \frac{16}{\pi d^3}\sqrt{M^2 + T^2}$$

$$= \frac{16}{\pi(0.11)^3}\sqrt{2678^2 + 8754^2}$$

$$= 35.0 \text{ MPa}$$

Max bending and torsional moment occurs at the inner bearing, when the shaft is at MCR and if the hydraulic pump was switched off.

$\frac{S_y}{2n} = \tau_{max}$ where S_y = yield stress

$$\text{FOS} \Rightarrow n = \frac{S_y}{2\tau_{max}} = \frac{300 \text{ MPa}}{2\times35.0\text{ MPa}}$$

$\boxed{n = 4.3}$ ✓

Further Calculations:

Minimum diameter for spline → 92mm

From ASME diameter formula:

$$n = \frac{d^3 \tau_p}{5.1\left[(C_m M)^2 + (C_t T)^2\right]^{1/2}}$$ using $\tau_p = 124$MPa

$$= \frac{(0.092)^3 \times (124\times10^6)}{5.1\left[(2\times2676)^2 + (1.5\times8754)^2\right]^{1/2}}$$

$\boxed{n_{(d=92mm)} = 1.3}$

Bearing Life Calcs

Equivalent Dynamic Load (P)

$F_r = 1.211$ kN $\quad F_a = 20$ kN $\quad X = 0.6 \quad Y = 1.07$

$P = XF_r + YF_a = 0.6\times1.211 + 1.07\times20 = 22.13$ kN

Basic Life (L_{10})

$C = 280 \quad P = 22.13 \quad p = 3$

$$L_{10} = \left(\frac{C}{P}\right)^p = \left(\frac{280}{22.13}\right)^3 = 2026 \text{ million revolutions}$$

$L_{10} = 23.45$ years

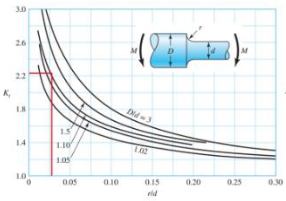
Lubrication Condition (k)

$\nu = 26.2 \text{ mm}^2/\text{s} \quad \nu_1 = 14.3 \text{ mm}^2/\text{s}$

$$k = \frac{\nu}{\nu_1} = \frac{26.2}{14.3} = 1.83$$

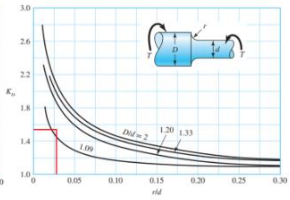
Shaft SCF Calculations

Shoulder in Bending



Round shaft with shoulder fillet in bending. $\sigma_0 = Mc/I$, where $c = d/2$ and $I = \pi d^4/64$.

Shoulder in Torsion



Round shaft with shoulder fillet in torsion. $\tau_0 = Tc/J$, where $c = d/2$ and $J = \pi d^4/32$.

$d = 110mm$
$D = 122mm$ ← Specified from SKF catalogue based on bearing selection

$D/d = 1.11$

$r = 3mm$

$r/d = 0.027$

From SCF diagrams:

$K_t \approx 2.23$

$K_{ts} \approx 1.55$

Bending at Shoulder

$I = \pi d^4/64$

$= \pi(0.11)^4/64 = 7.19 \times 10^{-6} m^4$

$\sigma_0 = \frac{Mc}{I}$

$= \frac{2678 \times 0.055}{7.19 \times 10^{-6}}$

$= 20.5 MPa$

$\sigma_{SCF} = \sigma_0 \cdot K_t$

$= 20.5 \times 2.23$

$= 45.7 MPa$

$n_{bending} = \frac{\sigma_y}{\sigma_{SCF}} = \frac{580MPa}{45.7MPa}$

$\boxed{n_{bending} = 12.7}$

Torsion at Shoulder

$J = \pi d^4/32$

$= \pi(0.11)^4/32 = 14.4 \times 10^{-6} m^4$

$\tau_0 = \frac{Tc}{J}$

$= \frac{8754 \times 0.055}{14.4 \times 10^{-6}}$

$= 33.4 MPa$

$\tau_{SCF} = \tau_0 \cdot K_{ts}$

$= 33.4 \times 1.55$

$= 51.8 MPa$

$\tau_{max} = \frac{\sigma_y}{\sqrt{3}}$ (Von Mises Criterion)

$= \frac{580}{\sqrt{3}}$

$= 334.9 MPa$

$n_{torsion} = \frac{\tau_{max}}{\tau_{SCF}}$

$= \frac{334.9}{51.8}$

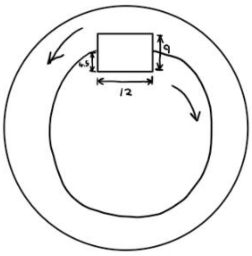
$\boxed{n_{torsion} = 6.5}$

Key Calculations

Key Material

Hot-Rolled AISI 1006 Low Carbon Steel

$\sigma_y = 165 MPa$ $\quad \tau_{max} = \frac{\sigma_y}{\sqrt{3}} = 95.3 MPa$



Key cross sectional dimensions:

$W = 12mm$ (width)
$H = 4.5mm^*$ (depth)

* Key is half in shaft, half in sprocket (total height is 9mm)

$D = 110mm$ (shaft diameter)

Key specified to fail at a torque of $2 \times$ max operating torque

$T_{fail} = 2 \times T_{max} = 2 \times 111.4Nm$

$= 222.8 Nm$

Determine key length based on shear and compressive failure modes:

Shear Failure:

$\tau_{key} = \frac{2T}{WLD}$

$\therefore L = \frac{2T}{\tau_{key} WD}$

$= \frac{2 \times 222.8}{95.3 \times 10^6 \times 0.012 \times 0.11}$

$= 3.54 mm$

Compressive Failure:

$\sigma_c = \frac{4T}{DLH}$

$\therefore L = \frac{4T}{\sigma_c DH}$

$= \frac{4 \times 222.8}{165 \times 10^6 \times 0.11 \times 0.0045}$

$= 10.9mm$

From these calculations a $\boxed{\text{length of } 11mm}$ is decided for the key.

FOS: (During standard operation)

Shearing

$\tau_{key} = \frac{2 \times 111.4}{0.012 \times 0.011 \times 0.11}$

$= 30.7 MPa$

$\boxed{n = \frac{95.3}{15.3} = 6.2}$

Compression

$\sigma_c = \frac{4 \times 111.4}{0.11 \times 0.011 \times 0.0045}$

$= 81.8 MPa$

$\boxed{n = \frac{165}{81.8} = 2.0}$

Key will fail in compression if the pump is overload by greater than twice its max operational capacity.

Clutch Calculations

DFGH

$T_{shaft} = 8754 Nm$

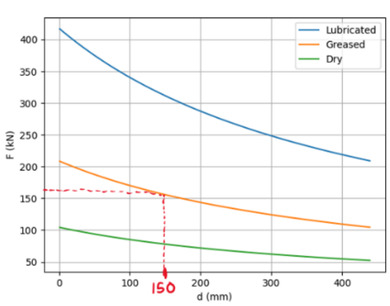
$T_{slip} = 9192 Nm$ (5% more)

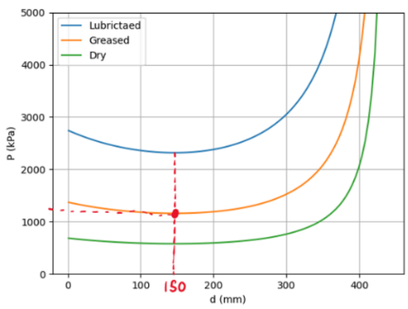
$T_w = \frac{Ff}{4}(D+d)$

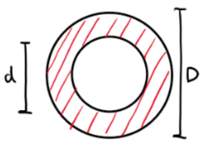
$F = \frac{4T_w}{f(D+d)}$

Choose $D = 440mm$ (max is 450mm), 4 clutch plates

vary d for different friction factors → see graphs







Area $= \frac{\pi D^2}{4} - \frac{\pi d^2}{4} = \frac{\pi}{4}(D^2 - d^2)$

$\sigma_{clutch} = \frac{F_{actuation}}{A}$

from graphs: choose $d = 150mm$ for $f = 0.1$ (greased steel on cast iron)

Assume uniform wear due to steel on cast iron:

Actuation force per plate (at slipping torque)

$F = \frac{4T_w}{f(D+d)n} = \frac{4 \times 9192}{0.1(0.44+0.150) \times 4} = \boxed{156kN}$

Maximum Contact Pressure per plate

$A = \frac{\pi}{4}(0.44^2 - 0.150^2) = 0.134 m^2$

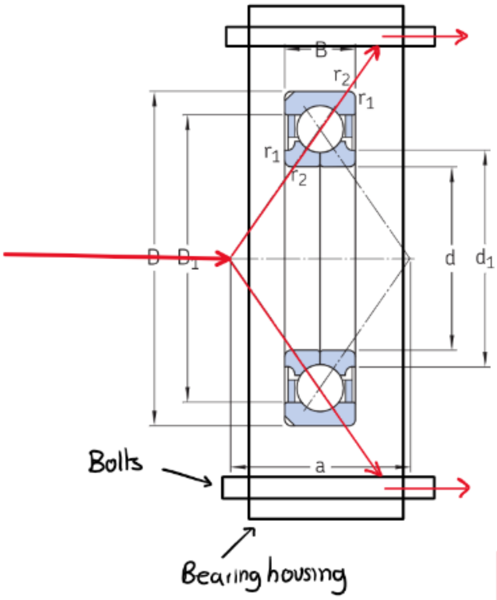
$P = \frac{F}{A} = \frac{156 \times 10^3}{0.134} = \boxed{1.16 MPa}$

Bearing Housing Calcs

Axial force on bearing housing:

$F = 20{,}000\ N$ (from F_{tan})

Loadpath Diagram



Bolt Stripping

There are four bolts which fix the bearing housing to the wall.

Each bolt takes 5,000 N of force.

Bolts are size M16.

Assuming 80% of bolt CSA takes the load,

$$A = 0.8 \times \frac{\pi \times (16\times10^{-3})^2}{4} = 0.161\times10^{-3}\ m^2$$

$$\sigma = \frac{F}{A} = \frac{5000}{0.161\times10^{-3}} = 31.1\ MPa \text{ per bolt.}$$

Grade A M16 Bolts (Medium Carbon Steel)

$\sigma_y = 660\ MPa$

$$\boxed{\therefore \text{ Bolt FOS}: \quad n = \frac{660}{31.1} = 21.2}$$

Bolt Shearing:

$$\tau_{max} \text{ of bolts} = \frac{660}{\sqrt{3}} = 381\ MPa \quad \text{(Von Mises)}$$

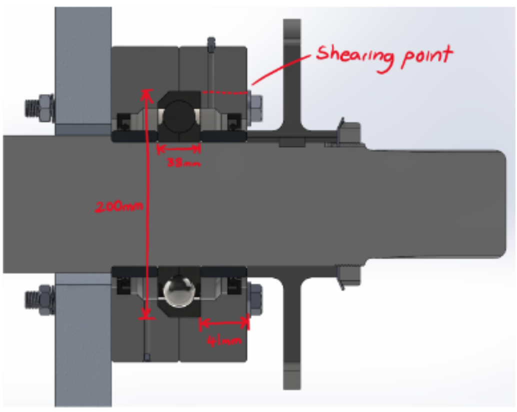
Radial reaction force at bearing = 1940 N

Max Secondary load from driveshaft = 5294 N

Combined max load: $P_{max} = \sqrt{1940^2 + 5294^2} = 5638\ N$

$$\sigma = \frac{P}{A} = \frac{5638/4}{0.161\times10^{-3}} = 8.8\ MPa \qquad \boxed{FOS = \frac{381}{8.8} = 43.3}$$

Housing Failure



bearing failure

$$A = D_{bearing} \times W_{bearing} = 0.2 \times 0.038 = 0.0076\ m^2$$

$$\sigma = \frac{F_{radial}}{A} = \frac{5638}{0.0076} = 0.74\ MPa$$

Shearing failure

$$A_{shear} = \pi D \times W = \pi(0.2)(0.041) = 0.026\ m^2$$

$$\tau = \frac{F_{axial}}{A_{shear}} = \frac{20{,}000}{0.026} = 0.77\ MPa$$

Housing Material is grey cast iron. Both bearing & shear stresses are small and FOS >> 100.

Speed and friction

The permissible operating temperature of rolling bearings imposes limits on the speed at which they can be operated. The operating temperature is determined, to a great extent, on the frictional heat generated in the bearing, except in machines where process heat is dominant.

An overview is provided in *Suitability of rolling bearings for industrial applications*, page 72, of the speed capability of various bearing types.

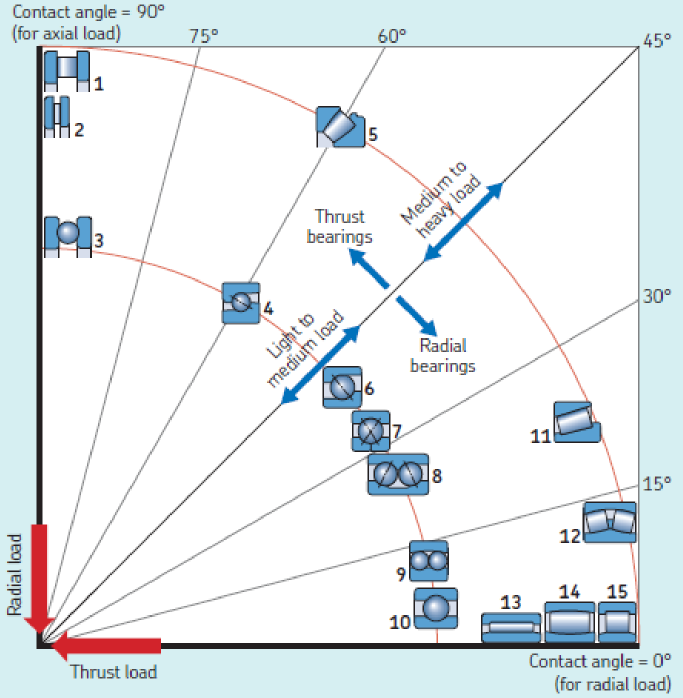
When selecting bearing type on the basis of operating speed, you should consider the following:

- Ball bearings have a lower frictional moment than same-sized roller bearings.
- Thrust bearings cannot accommodate speeds as high as same-sized radial bearings.
- Single row bearing types typically generate low frictional heat and are therefore more suitable for high-speed operation than double or multi-row bearings.
- Bearings with rolling elements made of ceramics (hybrid bearings) accommodate higher speeds than their all-steel equivalents.

Diagram 1

Contact angles of various bearing types

1 Cylindrical roller thrust bearing
2 Needle roller thrust bearing
3 Thrust ball bearing
4 Angular contact thrust ball bearing
5 Spherical roller thrust bearing
6 Single row angular contact ball bearing
7 Four-point contact ball bearing
8 Double row angular contact ball bearing
9 Self-aligning ball bearing
10 Deep groove ball bearing
The contact angle depends on load and clearance.
11 Tapered roller bearing
12 Spherical roller bearing
13 Needle roller bearing
14 CARB toroidal roller bearing
15 Cylindrical roller bearings



Suitability of rolling bearings for industrial applications

Symbols

+++ excellent
++ good
+ fair
- poor
– – unsuitable

⇔ double direction
← single direction
□ non-locating displacement on the seat
■ non-locating displacement within the bearing
✓ yes
✗ no

Bearing type		Load carrying capability: Radial load	Load carrying capability: Axial load	Load carrying capability: Moment load	Misalignment: Static misalignment	Misalignment: Dynamic misalignment (few tenths of a degree)
Deep groove ball bearings	A, B	+	+ ⇔	A –, B +	–	– –
Insert bearings	A, B, C	+	+ ⇔	– –	++	– –
Angular contact ball bearings, single row		+[1]	++ ←	– –	–	– –
matched single row	A, B, C	A, B ++ C ++[1]	A, B ++ ⇔ C ++ ←	A ++, B + C – –	A, C – –, B –	– –
double row	A, B	++	++ ⇔	++	– –	– –
four-point contact		+[1]	++ ⇔	– –	– –	– –
Self-aligning ball bearings		+	–	– –	+++	+[2]
Cylindrical roller bearings, with cage	A, B	++	– –	– –	–	– –
	A, B, C, D	++	A, B + ← C, D + ⇔	– –	–	– –
full complement, single row	A, B	+++	+ ←	– –	–	– –
full complement, double row	A, B, C, D	+++	A – –, B + ← C + ⇔	– –	–	– –
Needle roller bearings, with steel rings	A, B, C	++	– –	– –	A, B – C ++	– –
assemblies / drawn cups	A, B, C	++	A, B – – C –	– –	–	– –
combined bearings	A, B, C	++	A –, B + C ++	– –	– –	– –
Tapered roller bearings, single row		+++[1]	++ ←	– –	–	– –
matched single row	A, B, C	A, B +++ C +++[1]	A, B ++ ⇔ C ++ ←	A +, B ++ C – –	A – B, C – –	– –
double row	A, B	+++	++ ⇔	A + B ++	A –, B – –	– –
Spherical roller bearings		+++	+ ⇔	– –	+++	+[2]
CARB toroidal roller bearings, with cage		+++	– –	–	++	–
full complement		+++	– –	–	++	–
Thrust ball bearings	A, B	– –	A + ← B + ⇔	– –	– –	– –
with sphered housing washer	A, B	– –	A + ← B + ⇔	– –	++	– –
Cylindrical roller thrust bearings		– –	++ ←	– –	– –	– –
Needle roller thrust beairngs		– –	++ ←	– –	– –	– –
Spherical roller thrust bearings		+[1]	+++ ←	– –	+++	+[2]

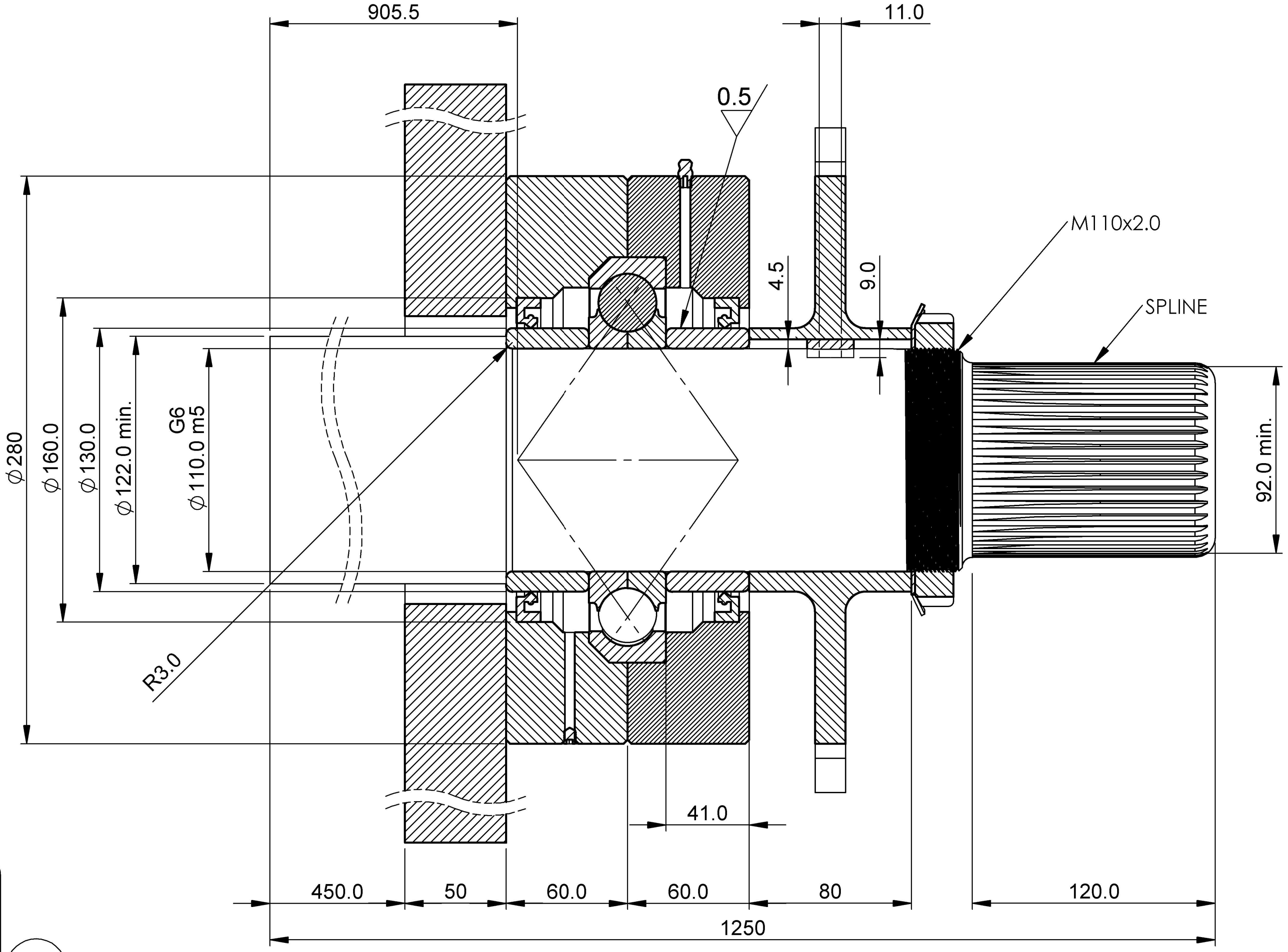
[1] Provided the F_a/F_r ratio requirement is met [2] Reduced misalignment angle – contact SKF [3] Depending on cage and axial load level

APPENDIX B

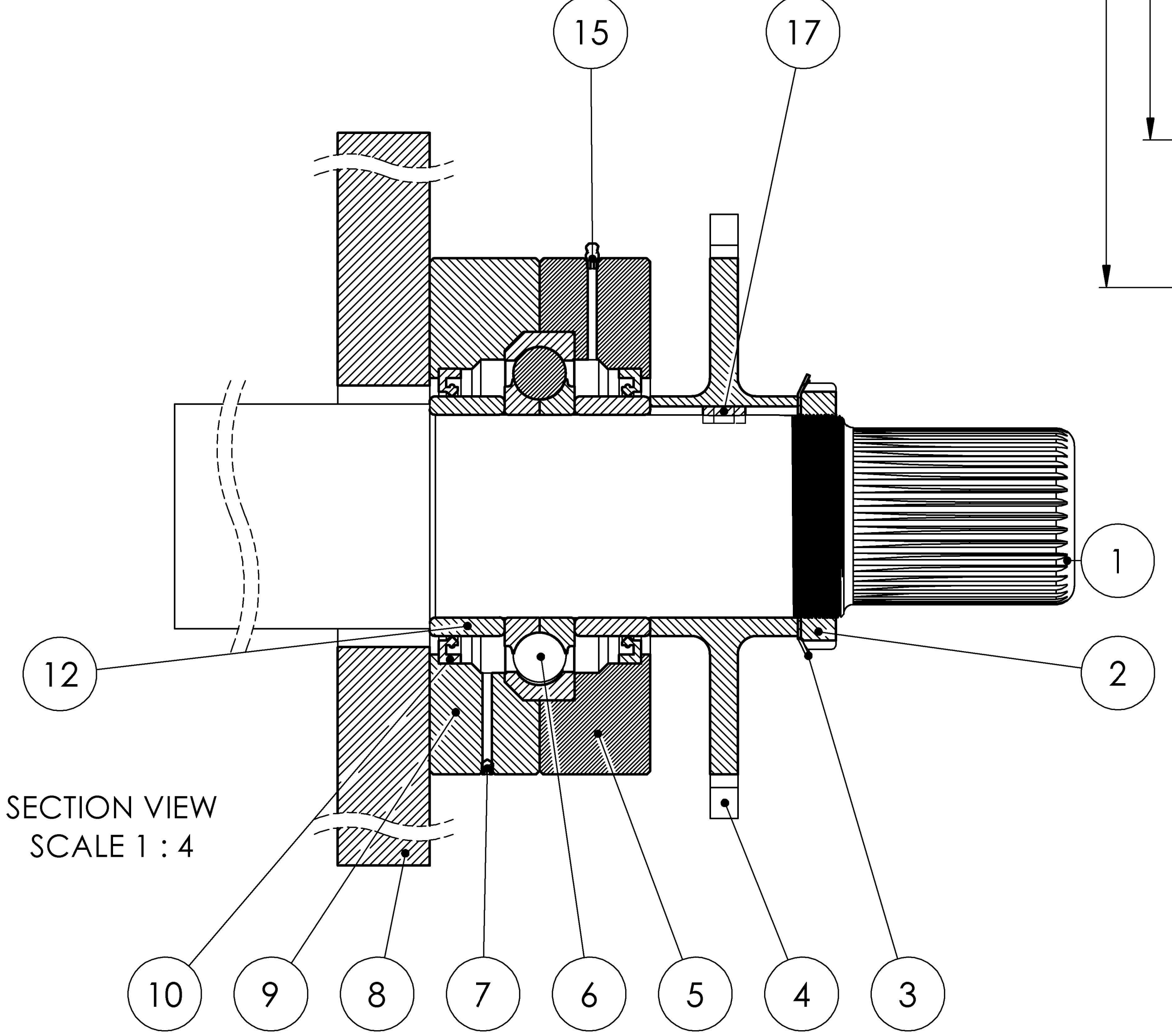
PART NO.	PART DESCRIPTION	QTY.
1	SHAFT	1
2	KM-22 LOCK NUT	1
3	MB-22 LOCK WASHER	1
4	SPROCKET (300 PCD)	1
5	OUTER BEARING HOUSING	1
6	QJ-222-N2MA BEARING	1
7	M6 GRUB SCREW	1
8	WALL	1
9	INNER BEARING HOUSING	1
10	HMS5-RG RADIAL SHAFT SEAL	2
11	M16 NUT	4
12	SEAL COUNTERFACE	2
13	M16 BOLT	4
14	M16 SPRING WASHER	4
15	M6 GREASE NIPPLE	1
16	M16 FLAT WASHER	4
17	KEY	1

NOTES - (UNLESS OTHERWISE SPECIFIED)

1. ALL DIMENSIONS IN MILLIMETERS.
2. GD&T AS PER ISO1101-2004.



SECTION VIEW
SCALE 1 : 3



SECTION VIEW
SCALE 1 : 4

THIRD ANGLE PROJECTION

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DEPARTMENT OF MECHANICAL ENGINEERING

MATERIAL: SEE PART DRAWINGS

FINISH: SEE PART DRAWINGS

BEARING HOUSING DESIGN

UNIVERSITY OF CANTERBURY
Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

TOLERANCE (UNLESS OTHERWISE SPECIFIED)
DECIMAL mm
X ± 1 .X ± .1 .XX ± .01 ANG. ± 1°

DESIGN	DWG NO.	REV
004	1	12

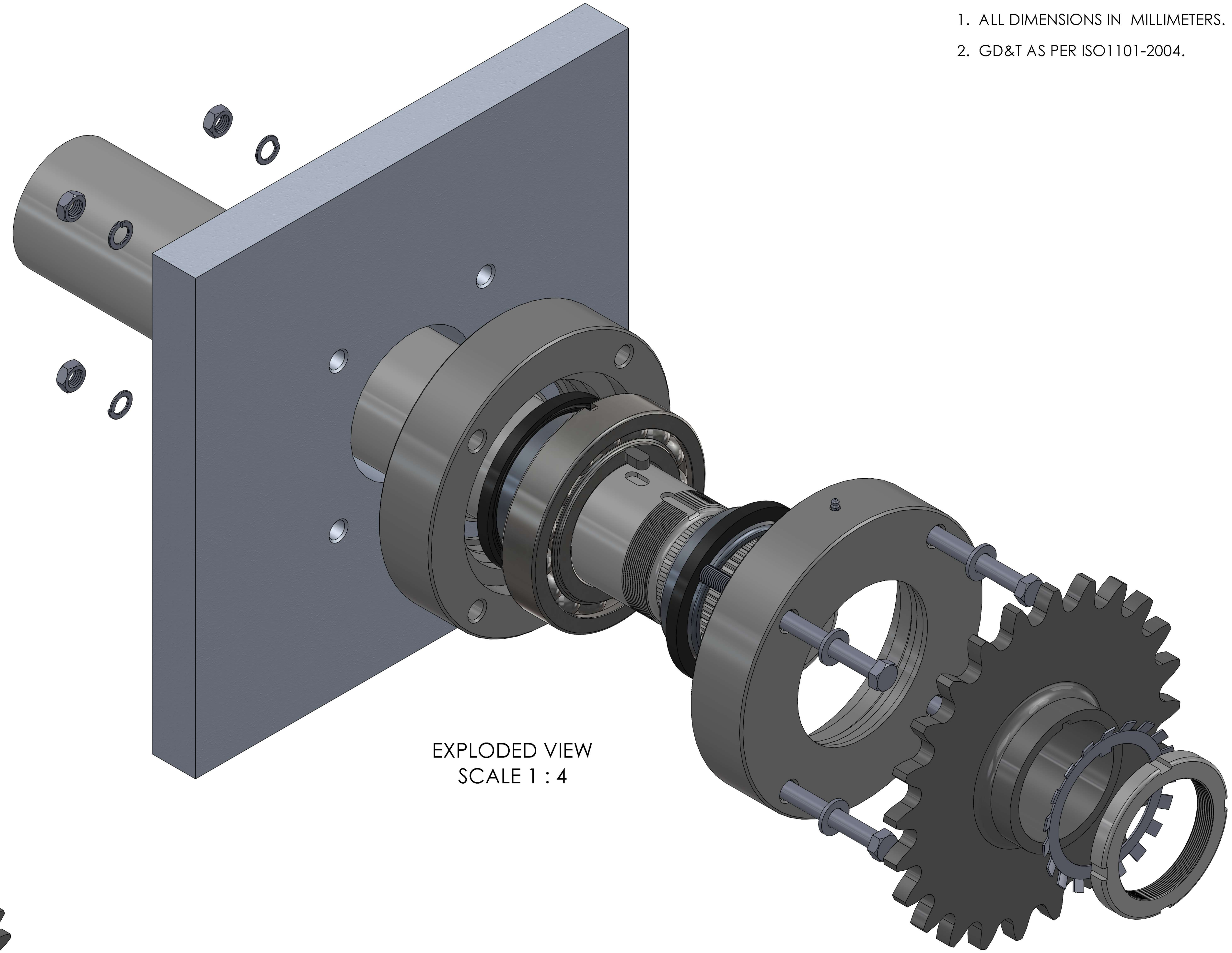
DRAWN: JAL244 - JLE193
PROJECT: ASSIGNMENT 2
SUPERVISOR: G.STILWELL
ISSUE DATE: 25/05/24
DRAWING NOT TO SCALE
SHEET: 1 OF 2

APPENDIX B

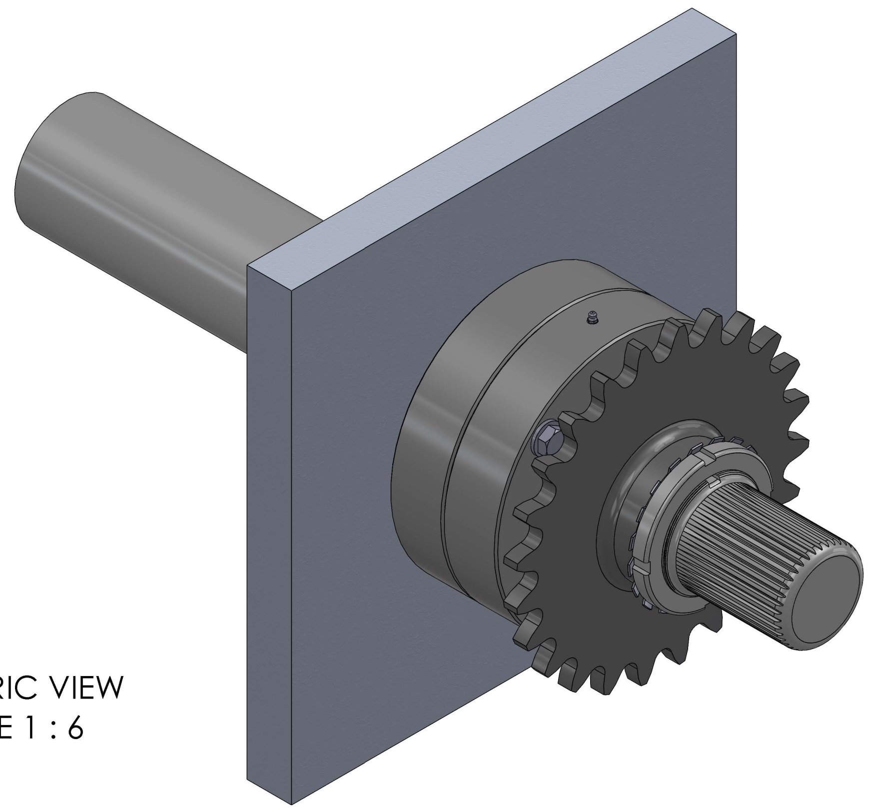
PART Description	QTY.
SHAFT	1
WALL	1
QJ-222-N2MA BEARING	1
SEAL COUNTERFACE	2
HMS5-RG RADIAL SHAFT SEAL	2
M6 GREASE NIPPLE	1
SPROCKET (300 PCD)	1
KEY	1
MB-22 LOCK WASHER	1
KM-22 LOCK NUT	1
M6 GRUB SCREW	1
M16 SPRING WASHER	4
M16 FLAT WASHER	4
M16 BOLT	4
M16 NUT	4
INNER BEARING HOUSING	1
OUTER BEARING HOUSING	1

NOTES - (UNLESS OTHERWISE SPECIFIED)

1. ALL DIMENSIONS IN MILLIMETERS.
2. GD&T AS PER ISO1101-2004.



EXPLODED VIEW
SCALE 1 : 4



ISOMETRIC VIEW
SCALE 1 : 6

THIRD ANGLE PROJECTION

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CHRISTCHURCH NEW ZEALAND

MATERIAL: SEE PART DRAWINGS

FINISH: SEE PART DRAWINGS

BEARING HOUSING DESIGN

TOLERANCE (UNLESS OTHERWISE SPECIFIED)
DECIMAL mm
X ± 1 .X ± .1 .XX ± .01 ANG. ± 1°

DESIGN	DWG NO.	REV
004	2	8

DRAWN: JAL244 - JLE193
PROJECT: ASSIGNMENT 2
SUPERVISOR: G.STILWELL
ISSUE DATE: 25/05/24
DRAWING NOT TO SCALE
SHEET: 2 OF 2